

Littlewood Polynomials, Flatness, and the Pandrosion Frobenius

Ivan Besevic

April 2026

Abstract

Littlewood polynomials $P(z) = \sum \varepsilon_i z^i$ ($\varepsilon_i \in \{-1, +1\}$) satisfy the universal Parseval identity $\frac{1}{2\pi} \int |P|^2 d\theta = n + 1$. The Rudin–Shapiro construction produces flat pairs with $|P|^2 + |Q|^2 = 2^{k+1}$. Via the Pandrosion Frobenius (Paper 75), all Littlewood polynomials reduce to the *same* polynomial $1 + z + \cdots + z^n$ over $\mathbb{F}_2$. We compute the Mahler measure distribution (Paper 59) and the merit factor for random Littlewood polynomials, and connect the Pandrosion field energy $\int |F_P|^2$ to the “derivative flatness” of ± 1 sequences.

1 Littlewood polynomials and Parseval

Definition 1.1. A *Littlewood polynomial* of degree n is $P(z) = \sum_{k=0}^n \varepsilon_k z^k$ with $\varepsilon_k \in \{-1, +1\}$.

Theorem 1.2 (Parseval, universal). *For every Littlewood polynomial of degree n :*

$$\boxed{\frac{1}{2\pi} \int_0^{2\pi} |P(e^{i\theta})|^2 d\theta = \sum_{k=0}^n |\varepsilon_k|^2 = n + 1.} \quad (1)$$

Verified at 10^{-16} precision for $n = 4, 8, 16, 32$.

2 Rudin–Shapiro: flat pairs

Proposition 2.1 (Rudin–Shapiro). *Define $P_0 = Q_0 = 1$ and:*

$$P_{k+1}(z) = P_k(z) + z^{2^k} Q_k(z), \quad Q_{k+1}(z) = P_k(z) - z^{2^k} Q_k(z). \quad (2)$$

Then P_k and Q_k are Littlewood polynomials of degree $2^k - 1$ satisfying:

$$\boxed{|P_k(z)|^2 + |Q_k(z)|^2 = 2^{k+1} \quad \text{on } S^1.} \quad (3)$$

Verified for $k = 2, \dots, 6$ (degrees 3 to 63) at precision 10^{-12} . The sup-norm ratio $\|P_k\|_\infty / \sqrt{n+1} \leq \sqrt{2}$.

3 Mahler measure distribution

n	Mean M	Median M	Min M	Max M
8	2.28	2.36	1.00	2.73
16	3.13	3.15	2.00	3.88
32	4.32	4.32	3.08	5.12

Computed over 1000 random samples per degree. The growth $M \sim c\sqrt{n}$ is consistent with the heuristic $M \approx \sqrt{n+1} \cdot e^{-\gamma/2}$.

4 Merit factor

Definition 4.1. The *merit factor* measures flatness on S^1 :

$$F(P) = \frac{(n+1)^2}{2 \frac{1}{2\pi} \int (|P|^2 - (n+1))^2 d\theta}. \quad (4)$$

The Rudin–Shapiro merit factor stabilizes near $F \approx 1.5$, while random Littlewood polynomials have $F \approx 0.5$ – 0.8 . The Littlewood conjecture (now disproved) asked whether F could grow without bound.

5 Frobenius universality over $\mathbb{F}_2$

Proposition 5.1. Over $\mathbb{F}_2$, every Littlewood polynomial of degree n reduces to:

$$P(z) \equiv 1 + z + z^2 + \cdots + z^n = \frac{z^{n+1} + 1}{z + 1} \pmod{2}, \quad (5)$$

since $\varepsilon_k \equiv 1 \pmod{2}$ for all $\varepsilon_k \in \{-1, +1\}$.

By the Frobenius (Paper 75): $(1 + z + \cdots + z^n)^2 = 1 + z^2 + z^4 + \cdots + z^{2n}$ over $\mathbb{F}_2$. All Littlewood polynomials are Frobenius-equivalent.

References

- [1] J. Littlewood, *Some Problems in Real and Complex Analysis*, Heath, 1968.
- [2] W. Rudin, H. Shapiro, *Extremal problems for polynomials*, Duke Math. J. **26**–**27** (1959–60).
- [3] I. Besevic, *Mahler measure via the Pandrosion field*, Pandrosion Dynamics **59** (2026).
- [4] I. Besevic, *Christol’s theorem and the Pandrosion Frobenius*, Pandrosion Dynamics **75** (2026).